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Project 29

Multi-class SVM with a Lagrangian Dual and Bundle Methods

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Chapter 1

Introduction

1.1 Description of the task

This report explores two different optimization algorithms applied to a machine learning model.

The ML model (M) is a Support Vector Machine-type approach for multi-class classification, with a choice of kernels, using the standard One-vs-Rest approach.

The first optimization algorithm (A1) is a dual approach applied to the dual formulation of the SVM. We reintroduce the weight vector w as an explicit variable, so that the representer identity $w = \sum_i \alpha_i y_i \phi(x_i)$ becomes a constraint of the problem, and we dualise *that* identity. The resulting Lagrangian relaxation is a linear program over a box with a single equality constraint — a continuous knapsack, solved exactly by sorting in $O(m \log m)$ — so the dual function is finite everywhere, genuinely non-differentiable, and defined on a space of dimension n for a linear kernel or m in general. It is maximised by a proximal bundle method, whose master problem is solved by an off-the-shelf QP solver, as the project statement allows.

The second optimization algorithm (A2) is a general-purpose convex QP solver applied to the standard SVM dual. No off-the-shelf solver is used elsewhere, save for the master problem of the bundle method.

Specific discussion is provided in Chapter 6 about if and how solving one training problem in the One-vs-Rest approach may provide useful solutions that help in solving the others.

Chapter 2

SVM Problem Formulation

The goal of this chapter is to describe the Support Vector Machine and to derive the Lagrangian dual formulation that defines the optimization task as a quadratic programming problem. The material is standard; it is included because Chapter 3 departs from it at one specific point, the stationarity condition (2.2), and that departure is easier to follow with the usual derivation in front of the reader.

2.1 Soft margin formulation

Let $\{(x_i, y_i)\}_{i=1}^m$ with $x_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$, and let $\phi : \mathbb{R}^n \to \mathcal{H}$ be a feature map into a Hilbert space with reproducing kernel $\kappa(x, x') = \langle \phi(x),\, \phi(x') \rangle$. Write $\phi_i := \phi(x_i)$.

A separating hyperplane $H(w, b) = \{x : \langle w,\, \phi(x) \rangle + b = 0\}$ classifies correctly with margin at least one when $y_i(\langle w,\, \phi_i \rangle + b) \geq 1$ for all i, and the geometric margin is then $1/\|w\|$, so maximising the margin means minimising $\|w\|$. Since such a hyperplane need not exist, each sample is given a slack $\xi_i \geq 0$ by which it may fall short of its own margin, penalised linearly with weight $C > 0$. This gives the soft-margin (C-)SVM:

$$\begin{aligned} \min_{w \in \mathcal{H},\, b \in \mathbb{R},\, \xi \in \mathbb{R}^m} \quad & \tfrac{1}{2}\|w\|^2 + C \sum_{i=1}^{m} \xi_i \\ \text{s.t.} \quad & y_i\big(\langle w,\, \phi_i \rangle + b\big) \;\geq\; 1 - \xi_i, \qquad i = 1, \ldots, m, \\ & \xi_i \;\geq\; 0, \qquad i = 1, \ldots, m. \end{aligned} \tag{P}$$

Here $\|w\|$ measures model complexity and C weighs the empirical loss against generalisation. Problem (P) is a convex QP, strictly convex in w but only convex in (b, ξ).

Eliminating ξ gives the equivalent unconstrained, non-smooth form

$$\min_{w,b} \ \tfrac{1}{2}\|w\|^2 + C\sum_{i=1}^{m}\big[1 - y_i(\langle w,\, \phi_i\rangle + b)\big]^+, \qquad [t]^+ := \max\{0, t\}, \tag{2.1}$$

which is the form in which the SVM appears in the course material as the motivating example for nondifferentiable optimization. We shall meet it again, unexpectedly, in Section 3.5.

2.2 Dual formulation

Attach $\alpha_i \geq 0$ to the margin constraints and $\mu_i \geq 0$ to $\xi_i \geq 0$:

$$L(w, b, \xi; \alpha, \mu) = \tfrac{1}{2}\|w\|^2 + C\mathbf{1}^\top\xi + \sum_{i=1}^{m}\alpha_i\big(1 - \xi_i - y_i(\langle w,\, \phi_i\rangle + b)\big) - \mu^\top\xi.$$

Stationarity in (w, b, ξ) gives the three conditions

$$w = \sum_{i=1}^{m}\alpha_i y_i \phi_i, \qquad \sum_{i=1}^{m}\alpha_i y_i = 0, \qquad \alpha_i + \mu_i = C. \tag{2.2}$$

The first is the *representer identity*; it is the hinge on which this whole project turns — Chapter 3 dualises precisely this condition.

Substituting the three conditions back into L makes every term collapse. Writing $K_{ij} = \kappa(x_i, x_j)$, $Y = \operatorname{diag}(y)$ and $Q = YKY$:

$$\begin{aligned}
\tfrac{1}{2}\|w\|^2 &= \tfrac{1}{2}\sum_{i,j}\alpha_i\alpha_j y_i y_j K_{ij} = \tfrac{1}{2}\alpha^\top Q\alpha, \\
-\sum_i \alpha_i y_i\langle w,\, \phi_i\rangle &= -\Big\langle w,\, \sum_i \alpha_i y_i\phi_i\Big\rangle = -\|w\|^2 = -\,\alpha^\top Q\alpha, \\
-\,b\sum_i \alpha_i y_i &= 0 \quad \text{(by the } b \text{ condition)}, \\
\sum_i \xi_i\big(C - \alpha_i - \mu_i\big) &= 0 \quad \text{(by the } \xi \text{ condition)},
\end{aligned}$$

leaving $\mathbf{1}^\top\alpha - \tfrac{1}{2}\alpha^\top Q\alpha$, to be maximised. Two remarks on how the constraints of the dual arise, because both are easy to state loosely:

- $\mu_i \geq 0$ and $\mu_i = C - \alpha_i$ together give $\alpha_i \leq C$, which with $\alpha_i \geq 0$ is the box $0 \leq \alpha \leq C\mathbf{1}$;

- b enters L only through the term $-b\sum_i \alpha_i y_i$, and b is *free*, so $\inf_b L = -\infty$ unless $\sum_i \alpha_i y_i = 0$. The b-stationarity condition therefore does not disappear when it is used to cancel that term: it survives as the constraint $y^\top \alpha = 0$.

Writing the result as a minimisation of the negative, the dual of (P) is the convex QP

$$\min_{\alpha \in \mathbb{R}^m} \ \tfrac{1}{2}\,\alpha^\top Q\,\alpha - \mathbf{1}^\top \alpha \qquad \text{s.t.} \qquad \alpha \in D := \big\{\alpha :\ y^\top \alpha = 0,\ \ 0 \le \alpha \le C\mathbf{1}\big\}. \tag{D}$$

D is a non-empty ($0 \in D$) compact polytope and $Q \succeq 0$, so (D) always has a solution. Slater's condition holds for (P), hence there is no duality gap and the KKT conditions

$$\begin{aligned} \alpha_i = 0 \quad &\Longrightarrow \quad y_i f(x_i) \ge 1, \\ 0 < \alpha_i < C \quad &\Longrightarrow \quad y_i f(x_i) = 1, \\ \alpha_i = C \quad &\Longrightarrow \quad y_i f(x_i) \le 1, \end{aligned} \tag{2.3}$$

characterise optimality, where $f(x) = \sum_i \alpha_i y_i \kappa(x_i, x) + b$. Samples with $\alpha_i > 0$ are the *support vectors*; those with $0 < \alpha_i < C$ are the *free* support vectors. Their number will turn out to control the cost of (A1); see Section 4.4.

2.3 Primal and dual variables

Given an optimal $\alpha^\star$ of (D), the primal variables come back as

$$w^\star = \sum_i \alpha_i^\star y_i \phi_i, \qquad \xi_i^\star = \big[\, 1 - y_i(\langle w^\star,\, \phi_i\rangle + b^\star)\,\big]^+,$$

the first directly from the representer identity and the second from primal feasibility of the margin constraint together with complementarity on $\xi_i \ge 0$ (note the factor y_i inside: ξ_i measures the violation of *the sample's own* margin, not of the hyperplane equation).

The bias $b^\star$ deserves more care. The textbook recipe reads $b^\star$ off any free support vector via (2.3) and averages over them for stability; but the average is only as good as the set of indices identified as free, so a solver that is merely close to optimal produces a b that degrades the measured primal–dual gap. The better observation is that b enters the Lagrangian only through $-b\sum_i \alpha_i y_i$, so b *is the multiplier of the equality constraint* $y^\top \alpha = 0$. Any algorithm that produces that multiplier explicitly can use it directly, with no support-vector bookkeeping at all. For an interior-point method it is one of the iterates; in our case the knapsack of Section 3.3 supplies it at every oracle call, as (3.3) proves by LP duality.

2.4 Kernel trick

Only inner products appear in (D), so ϕ need never be formed. The kernels used in this report are

- linear: $\kappa(x, x') = x^\top x'$;
- polynomial: $\kappa(x, x') = (\gamma\, x^\top x' + c_0)^d$, $\gamma > 0$;
- Gaussian (RBF): $\kappa(x, x') = \exp(-\gamma \|x - x'\|^2)$, $\gamma > 0$.

Unlike the linear case, for a nonlinear kernel $\mathcal{H}$ may be infinite-dimensional; Section 4.3 explains how the dual variable of (A1) is represented in that case.

Chapter 3

The Lagrangian Dual

3.1 Which constraints to dualise

For a Lagrangian dual approach to be worth building, the relaxation must satisfy:

(R1) the relaxed problem must be *substantially easier* than the original — ideally closed-form or combinatorial;

(R2) the dual must be *multi-dimensional*, otherwise a one-dimensional search solves it and bundle machinery is pointless;

(R3) the dual should be *genuinely non-differentiable*, otherwise a smooth method is preferable.

Relaxing the single equality $y^\top \alpha = 0$ in (D) violates (R1) and (R2): what is left is

$$\min\Big\{ \tfrac{1}{2}\alpha^\top Q\alpha - \mathbf{1}^\top\alpha + \theta\, y^\top\alpha \;:\; 0 \le \alpha \le C\mathbf{1} \Big\},$$

a box-constrained QP with the *same dense Hessian* — the unbiased SVM, no easier than the original — and the dual is univariate. Relaxing the box $\alpha \le C\mathbf{1}$ fares no better: the relaxation is again a QP in Q.

The reason is structural: in (D) *no constraint is the difficulty.* The difficulty is Q itself, a dense $m \times m$ matrix. So the object to dualise is not a constraint of (D) at all — it is the identity that *created* Q.

3.2 Dualising the representer identity

Q appears in (D) only because w was eliminated through the representer identity (2.2). Put w back:

$$\min_{\alpha,\, w} \quad \tfrac{1}{2}\|w\|^2 - \mathbf{1}^\top\alpha \qquad \text{s.t.} \qquad w = \sum_{i=1}^{m} \alpha_i y_i \phi_i, \qquad \alpha \in D. \tag{D$'$}$$

Substituting the constraint recovers (D) exactly, so (D$'$) and (D) are the same problem. Now dualise *only* the linking identity, with a free multiplier $\lambda \in \mathcal{H}$:

$$L(\alpha, w; \lambda) = \tfrac{1}{2}\|w\|^2 - \mathbf{1}^\top\alpha + \Big\langle \lambda,\ w - \sum_i \alpha_i y_i \phi_i \Big\rangle.$$

The two groups of variables separate. Minimising over $w \in \mathcal{H}$ is unconstrained and gives $w(\lambda) = -\lambda$ with value $-\tfrac{1}{2}\|\lambda\|^2$; what remains is linear in α over D. Hence

$$\boxed{\psi(\lambda) \;=\; -\tfrac{1}{2}\|\lambda\|^2 \;+\; \min_{\alpha \in D}\; c(\lambda)^\top\alpha, \qquad c_i(\lambda) = -\big(1 + y_i\langle \lambda,\, \phi_i\rangle\big),} \tag{3.1}$$

and the Lagrangian dual of (D$'$) is $\max_{\lambda\in\mathcal{H}} \psi(\lambda)$.

In the vocabulary of partial Lagrangian relaxation, the representer identity is the *complicating* constraint — it is what couples w to α and so what creates the dense Q — while $\alpha \in D$ are the *easy* constraints, kept in the relaxed problem because they are exactly what makes it a knapsack. The usual warning that $\psi(\lambda)$ may be $-\infty$ for some λ, and that one then has to neutralise the offending directions of zero curvature, does not apply here: D is compact, so the relaxed problem always has a finite optimum and ψ is finite on all of $\mathcal{H}$.

3.3 The relaxed problem is a continuous knapsack

The inner problem in (3.1) is

$$\min\ c^\top\alpha \qquad \text{s.t.} \qquad y^\top\alpha = 0, \quad 0 \le \alpha \le C\mathbf{1}, \tag{KP}$$

a linear program over a box with a *single* equality constraint: the classical *continuous knapsack* problem. Relaxing that one equality with a scalar θ makes the box minimiser explicit,

$$\alpha_i(\theta) = \begin{cases} C, & c_i - \theta y_i < 0, \\ 0, & c_i - \theta y_i > 0, \\ \text{any value in } [0, C], & c_i - \theta y_i = 0, \end{cases}$$

and $h(\theta) := y^\top \alpha(\theta)$ is non-decreasing, so a $\theta^\star$ with $h(\theta^\star) = 0$ exists and is found by sorting the m breakpoints $y_i c_i$ and scanning (or by median search in $O(m)$ expected time). Splitting the tied components to make the equality hold exactly produces an α that minimises the Lagrangian of (KP) *and* is feasible, hence optimal. Forming c costs one kernel product; the rest costs $O(m \log m)$. This settles (R1): a dense QP has been replaced by a sort.

The same structure — relax a single equality, obtain a separable problem whose solution is piecewise linear in the multiplier with one breakpoint per variable — appears as a worked exercise in the course material for the projection onto a hyperplane, and (KP) is its bounded-variable analogue.

3.4 Properties of the dual function

Proposition 3.1 (finiteness)**.** *D is a non-empty compact polytope, so the minimum in* (3.1) *is attained and finite for every λ. Hence $\psi : \mathcal{H} \to \mathbb{R}$ is finite everywhere, concave and upper semicontinuous.*

This is unusually convenient: no feasibility ("infinite") cuts are ever needed, and every oracle call returns a valid lower bound on the optimal value.

Proposition 3.2 (structure and non-differentiability)**.** *Write*

$$\psi = \psi_{\mathrm{sm}} + \psi_{\mathrm{pl}}, \qquad \psi_{\mathrm{sm}}(\lambda) = -\tfrac{1}{2}\|\lambda\|^2, \quad \psi_{\mathrm{pl}}(\lambda) = \min_{\alpha \in D} c(\lambda)^\top \alpha.$$

Then ψ_{sm} is smooth and known in closed form, and ψ_{pl} is a minimum of finitely many affine functions of λ (one per vertex of D), hence concave polyhedral. ψ fails to be differentiable exactly where (KP) *has multiple optima, i.e. where some reduced cost vanishes, $c_i - \theta^\star y_i = 0$.*

Unwinding that condition with $w = -\lambda$ gives $y_i \langle w, \phi_i \rangle + \theta^\star y_i = 1$: the kinks of ψ sit *exactly* on the support-vector condition $y_i f(x_i) = 1$ of (2.3). Since any non-degenerate SVM has free support vectors, the maximiser of ψ *is* a point of non-differentiability. This settles (R3): the non-smoothness is not incidental, it is the solution.

Proposition 3.3 (no duality gap)**.** (D′) *is convex, its coupling constraint is affine and D is compact; therefore* $\max_\lambda \psi(\lambda) = v(\mathrm{D}') = v(\mathrm{D})$*, and the maximiser is $\lambda^\star = -w^\star$.*

The multiplier of the relaxed identity is thus, up to sign, the primal weight vector itself. A supergradient of ψ at λ is the violation of the relaxed constraint,

$$g(\lambda) = w(\lambda) - \sum_i \alpha_i(\lambda) y_i \phi_i = -\lambda - \sum_i \alpha_i(\lambda) y_i \phi_i, \tag{3.2}$$

available at no extra cost from the oracle. (R2) holds as well: λ ranges over $\mathcal{H}$, of dimension n for the linear kernel and m after the reduction of Section 4.3 in general.

3.5 What this dual actually is

It is worth identifying $\max_\lambda \psi$ in closed form. Substituting $w = -\lambda$ in (3.1),

$$\max_{\lambda} \psi(\lambda) \iff -\min_{w} \Big\{ \tfrac{1}{2}\|w\|^2 + \max_{\alpha \in D} \sum_i \alpha_i \big(1 - y_i \langle w,\, \phi_i \rangle\big) \Big\},$$

and dualising the single equality inside the inner maximisation with a scalar θ gives, by LP duality,

$$\max_{\alpha \in D} \sum_i \alpha_i \big(1 - y_i g_i\big) = \min_{\theta} \; C \sum_i \big[1 - y_i (g_i + \theta)\big]^+, \qquad g_i := \langle w,\, \phi_i \rangle. \tag{3.3}$$

So the Lagrangian dual of (D′) is the hinge-loss primal (2.1), and *the multiplier θ of the knapsack equality is the SVM bias b* — so b falls out of the oracle for free. That the dual of the dual is the primal is of course biconjugacy; the next section says what is nevertheless gained.

3.6 Relation to the direct nonsmooth formulation

The SVM appears in the course material on nondifferentiable unconstrained optimization as the motivating example, in the form

$$\min_{w,b} \quad \|w\|^2 + C \sum_{i \in I} \max\big\{1 - y_i(\langle w,\, x_i \rangle - b),\, 0\big\},$$

whose nondifferentiability is precisely what motivates bundle methods. Since Section 3.5 shows that $\max_\lambda \psi$ is that same function, we must say clearly what dualising the representer identity adds over simply running a bundle method on it — which is also what the published cutting-plane methods for regularised risk minimisation do (Joachims 2006; Teo et al. 2010; Franc & Sonnenburg 2008).

The answer is *not* that the bias makes a direct approach impossible: $f(w, b)$ is convex in (w, b) jointly and a subgradient in both variables is immediate, so one can perfectly well put b in the bundle. The answer is that

our relaxation minimises over b *exactly*, at every oracle call, instead of approximating the b direction with cutting planes. The knapsack multiplier θ of (3.3) *is* that exact partial minimisation, and the sort of Section 3.3 is how it is computed. Equivalently: we bundle $g(w) = \min_b f(w, b)$ rather than $f(w, b)$, and g is the better-conditioned object because one of its directions has been eliminated in closed form. The measured effect is reported in Section 7.2.

We claim exactly this much: the family of cuts and the function being optimised are the same as in the direct approach, and what relaxing the representer identity adds is (i) the exact elimination of b at every oracle call, (ii) the SVM dual variables through the aggregated solution of Section 4.2, and (iii) a two-sided bound on $v(\mathrm{D})$ rather than a bound on one's own objective only.

Chapter 4

The Bundle Method

4.1 Model and master problem

Because $-\frac{1}{2}\|\lambda\|^2$ is known in closed form, we linearise only the polyhedral part — a *generalized* (structured) bundle method. Each oracle call returns some $\alpha^j \in D$, which gives the affine upper model

$$\psi_{\mathrm{pl}}(\lambda) \;\le\; c(\lambda)^\top \alpha^j = -\,\mathbf{1}^\top \alpha^j - \langle \lambda,\, \textstyle\sum_i \alpha_i^j y_i \phi_i \rangle =: -a_j - \langle \lambda,\, b_j \rangle.$$

With a stability centre $\bar\lambda$ and proximal parameter $t > 0$ the master is

$$\max_{\lambda \in \mathcal{H}} \;\; -\tfrac{1}{2}\|\lambda\|^2 + \min_{j=1,\dots,J} \big\{ -a_j - \langle \lambda,\, b_j \rangle \big\} - \tfrac{1}{2t}\|\lambda - \bar\lambda\|^2. \tag{4.1}$$

Exchanging min and max and eliminating λ gives its dual, a QP over the unit simplex Δ^J:

$$\min_{\vartheta \in \Delta^J} \;\; \frac{1}{2\rho} \Big\| \tfrac{\bar\lambda}{t} - \textstyle\sum_j \vartheta_j b_j \Big\|^2 - \;\textstyle\sum_j \vartheta_j a_j, \qquad \rho := 1 + \tfrac{1}{t}, \tag{4.2}$$

after which $\lambda^{\mathrm{new}} = \big(\bar\lambda/t - \sum_j \vartheta_j b_j\big)/\rho$. Only the $J \times J$ Gram matrix $G_{jl} = \langle b_j,\, b_l \rangle$ and the vector $p_j = \langle b_j,\, \bar\lambda \rangle$ are needed; both are maintained incrementally, so a master costs $O(J^2)$ plus one kernel product per new cut. We solve it with an off-the-shelf QP solver, which the project statement allows.

4.2 Primal recovery and Dantzig–Wolfe

Let $\hat\alpha := \sum_j \vartheta_j \alpha^j$ be the *aggregated solution* built from the master problem's dual multipliers. Since $\vartheta \in \Delta^J$ and D is convex, $\hat\alpha \in D$ — it is a *feasible*

solution of (D) at every iteration. Moreover $\sum_j \vartheta_j b_j = \Phi^\top Y \hat{\alpha}$ and $\sum_j \vartheta_j a_j = \mathbf{1}^\top \hat{\alpha}$, so for $t \to \infty$ (4.2) becomes

$$\min_{\vartheta \in \Delta^J} \ \tfrac{1}{2}\, \hat{\alpha}^\top Q \hat{\alpha} - \mathbf{1}^\top \hat{\alpha},$$

i.e. *the SVM dual* (D) *restricted to* $\operatorname{conv}\{\alpha^1, \ldots, \alpha^J\} \subseteq D$. The method is therefore exactly Dantzig–Wolfe decomposition of (D) with (KP) as pricing problem, and bundle, cutting planes on (2.1) and column generation on (D) are three views of one algorithm. Two practical consequences:

- the master returns the primal solution $\hat{\alpha} \to \alpha^\star$, so nothing extra is needed to recover the classifier;
- $\psi(\lambda) \le v(\mathrm{D}) \le \tfrac{1}{2}\hat{\alpha}^\top Q \hat{\alpha} - \mathbf{1}^\top \hat{\alpha}$ at *every* iteration, a **certified gap** — which decomposition algorithms for the SVM such as SMO do not provide.

4.3 Kernelised form

For a general kernel $\mathcal{H}$ may be infinite-dimensional, so write $\lambda = \sum_j \beta_j \phi_j$. This is without loss of generality: a component of λ orthogonal to $\operatorname{span}\{\phi_i\}$ leaves $c(\lambda)$ unchanged and only increases $\|\lambda\|^2$, so it is zero at the optimum. Then $\langle \lambda, \phi_i \rangle = (K\beta)_i$ and $\|\lambda\|^2 = \beta^\top K \beta$, and one oracle call costs one matrix–vector product with K plus a sort. Note that the correct stabilising term is the *RKHS* norm $\|\lambda - \bar{\lambda}\|^2 = (\beta - \bar{\beta})^\top K (\beta - \bar{\beta})$ rather than the Euclidean $\|\beta - \bar{\beta}\|^2$; it is the K-metric that makes (4.2) come out in the form above.

4.4 Cost, and how many cuts are needed

Per iteration: one oracle call ($O(m^2)$ for the kernel product, $O(m \log m)$ for the sort) plus one master ($O(J^2)$ in the QP, $O(Jm)$ to update G). The interesting question is how large J must grow. The oracle returns essentially *vertices* of D, that is $0/C$ vectors, and $\hat{\alpha}$ is a convex combination of them: a component of $\alpha^\star$ sitting at a bound is produced by a single vertex, but every *free* support vector $0 < \alpha_i^\star < C$ must be produced by mixing vertices that disagree on i. We therefore expect, and measure in Section 7.3, that the number of cuts the method needs is governed by the number of free support vectors, and that truncating the bundle below that number makes the gap stall — the classical Dantzig–Wolfe tailing-off.

Chapter 5

The General-Purpose Solver

For (A2) the dual (D) is handed as a dense convex QP to an off-the-shelf interior-point solver. Mapping it onto the generic QP interface $\min\{\frac{1}{2}z^\top Hz + f^\top z : A_{\text{eq}}z = b_{\text{eq}},\ l \le z \le u\}$ requires only

$$H = Q, \qquad f = -\mathbf{1}, \qquad A_{\text{eq}} = y^\top, \qquad b_{\text{eq}} = 0, \qquad l = 0, \qquad u = C\mathbf{1},$$

and the bias is read off as the multiplier of the single equality row, exactly as in Section 2.3. This is the reference both for solution quality — we verify that (A1) and (A2) return the same $\alpha^\star$, b and test accuracy — and for running time.

Chapter 6

Multiclass SVM

6.1 One-vs-Rest approach

The SVM is a binary classifier. With q classes and labels $c_i \in \{1, \dots, q\}$, One-vs-Rest solves q independent binary problems: for class k set

$$y_i^k = \begin{cases} +1, & c_i = k \\ -1, & c_i \neq k \end{cases}$$

and solve (P)/(D) with $y = y^k$, obtaining $f^k(x) = \sum_i \alpha_i^k y_i^k \kappa(x_i, x) + b^k$. A new point is assigned to $\arg\max_k f^k(x)$. It constructs q models, against $q(q-1)/2$ for One-vs-One.

Three observations matter for what follows:

1. The q problems share *the same* kernel matrix K; they differ only through the label vector, since $Q^k = Y^k K Y^k$ and D^k depends on k only through the single equality $(y^k)^\top \alpha = 0$. Q^k is never formed.

2. The classifiers are trained independently but compared at prediction time on a common scale that nothing in the training enforces; this produces the familiar One-vs-Rest pathologies, ties ($f^k(x) > 0$ for two k) and rejections ($f^k(x) < 0$ for all k).

3. Each binary problem is unbalanced, roughly $1 : (q-1)$.

6.2 Does solving one problem help the others?

The project statement asks this explicitly. The q problems differ only through y^k, so there are three candidate mechanisms, and they are not equally useful.

The kernel matrix. K is computed once and shared; $Q^k = Y^k K Y^k$ is never formed, since the oracle only ever needs $K\beta$. Better still, the q oracles can be run in lockstep so that their q matrix–vector products become one BLAS-3 product $K\,[\beta^1 \cdots \beta^q]$. This is a genuine, structural saving of a factor q in the dominant cost.

Warm-starting the multiplier. Since $\lambda^\star = -w^\star$, warm-starting class $k{+}1$ at class k's optimum amounts to assuming the two separating hyperplanes are similar. They are not: the positive class changes completely between the two problems.

Transferring cuts. Any point of D^l generates a valid cut for class l, and the D^k differ only through the equality. Writing P_k, P_l for the two positive blocks and R for the rest, $\alpha \in D^k$ satisfies $\alpha(P_k) = \alpha(P_l) + \alpha(R)$, so $(y^l)^\top \alpha = -2\,\alpha(R)$: the imbalance is known in closed form and can be repaired in $O(m)$ by moving mass onto P_l, *preserving the support pattern* of the transferred cut. This costs no oracle call and no kernel product.

Section 7.4 reports what each of the three is actually worth.